



A Left-Turn Driving Aid Using Projected Oncoming Vehicle Paths with Augmented Reality

Cuong Tran, Karlin Bark, Victor Ng-Thow-Hing
Honda Research Institute, USA
425 National Avenue, Mountain View, CA, USA
ctran, kbark, vng@honda-ri.com

ABSTRACT

Making left turns across oncoming traffic without a protected left-turn signal is a significant safety concern at intersections. In a left turn situation, the driver typically does not have the right of way and must determine when to initiate the turn maneuver safely. It has been reported that a driver's inability to correctly judge the velocity and time gap of the oncoming vehicles is a major cause of left turn crashes. Although the position and velocity of surrounding vehicles is available using camera and laser based vehicle detection and tracking, methods on how to effectively communicate such information to help the driver have been relatively under-explored. In this paper, we describe a left turn aid that displays a 3 second projected path of the oncoming vehicle in the driver's environment with a 3D Head-Up Display (3D-HUD). Utilizing the abilities of our 3D-HUD to show the projected path in Augmented Reality (AR) could help increase driver intuition and alleviate visual distraction as compared to other possible non-AR solutions. Through an iterative process utilizing early user feedback, the design of the left turn aid was refined to interfere less with the driver view and be more effective. A pilot study has been designed for a driving simulation environment and can be used to evaluate the potential of the proposed AR left turn aid in helping the driver be more cautious or efficient when turning left.

Keywords

Advanced Driver Assistance System, Head-Up-Display, Augmented Reality, Left Turn Assistance

1. INTRODUCTION

Left turn accidents are considered to be one of the most dangerous intersection accidents with high potential of injury and vehicle damage [4]. A common left turn crash scenario is the Left Turn Across Path/Opposing Direction (LTAP/OD), where the driver makes a left turn across the path of a vehicle approaching from the opposite direction.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from Permissions@acm.org. AutomotiveUI '13, October 28 - 30 2013, Eindhoven, Netherlands Copyright 2013 ACM 978-1-4503-2478-6/13/10...\$15.00. <http://dx.doi.org/10.1145/2516540.2516581>

This accounts for 27.3% of all US intersection-related crashes [17]. Causes of these crashes are the driver's inability to correctly judge safe time gaps, inability to judge the speed of oncoming vehicles, and failure to see the oncoming vehicles (e.g., looked but did not see, or driver's view was obstructed by an opposing vehicle) [17]. Advancements in vehicle detection and tracking research [25], as well as commercial product developments [11], have made it possible to detect and measure the position and velocity of oncoming vehicles during a left turn scenario. However, methods to communicate this information to the driver for use in a left turn scenario with minimal distraction and cognitive load have not been developed.

In this paper, we present a left turn driving aid that utilizes a 3D Augmented Reality Head-Up Display (3D AR-HUD) that displays a 3 second projected path of the oncoming vehicle in the driver's main field of view. The advantages of using a HUD have been outlined in Ablabmeier et al. [1], where they demonstrated that communicating information through a HUD versus a head-down display (HDD) reduced gaze retention periods, resulting in less visual distraction. Moreover, our 3D AR-HUD also has an advantage over more traditional contact analog HUDs in that it can display information in augmented reality which is directly in context of the driver's view in the real world. This can help reduce the driver's cognitive distraction since there is no need to translate information between the 3D AR-HUD view and the driver's primary view [2]. Compared to commonly known advanced driver assistance systems (ADAS) such as pedestrian or vehicle detection, which typically inform the driver about the existence and location of the surrounding objects at that instance of time, the proposed projected path also aims to increase the driver's awareness of the situation towards the near future rather than just the present state.

Since 1991, the National Highway Traffic Safety Administration (NHTSA) has conducted research on driver distraction involving visual, biomechanical, and cognitive distraction (e.g., eyes off road, hands off wheel, and mind off driving), and have developed methods to assess the extent to which in-vehicle technologies may contribute to crashes [22]. However, there are concerns about possible driver distraction due to the compatibility of certain in-vehicle technologies with the driving task. To address these concerns, we have developed a pilot study designed to test the effectiveness of our driving aid and methods.

2. RELATED RESEARCH STUDIES

2.1 Left Turn Gap Acceptance

Vehicular crashes involving intersections represent approximately 36% of all crashes from July 3, 2005 to December 21, 2007 [5]. 49% of LTAP/OD crashes were caused by drivers unaware of the oncoming vehicle, and 30% were caused by drivers who misjudged the velocity and acceptance gap of the oncoming vehicle [6]. Research on left turn driving behavior shows that there are numerous factors that influence a driver's gap acceptance. Yan et al. [29] noted differences in gap acceptance between driver age and gender: For drivers 55 years and older, the misjudgment of gap or oncoming speed is a major cause of left turn crashes [5], reflecting the reduced ability of older drivers to integrate time based on speed and distance. As a result, older drivers compensate by accepting larger gaps between themselves and the oncoming cars. Gender differences in gap acceptance have also been observed, where female drivers tend to be more conservative drivers, accepting larger gaps than male drivers. Male drivers tend to overestimate their perceptual-motor skills [14].

Drivers also tend to use unrelated or unreliable factors for making decisions about safe gap acceptance. At higher road speeds, drivers are more likely to accept smaller gap distances and exhibit more sensitivity to distance over vehicle speed [29]. Caird and Hancock [3] demonstrated that a driver's perception of the oncoming vehicle's arrival time is affected by the size of the vehicle. In their work, participants were more conservative in their arrival time estimates as physical size of vehicles increased.

Drivers have also exhibited wide variance on what type of gap lengths they would accept, ranging anywhere from 2 to 12 seconds [21]. Consequently, collision avoidance systems located at intersections that give a warning at a fixed gap length may elicit different responses from drivers, as they compare the warning system against their own personal preferences. Drivers may end up choosing to ignore the warning systems.

The wide range of factors that impact the driver's perception of the oncoming vehicle's position and speed highlight the need for a system that can improve all drivers' ability to judge the situation and reduce the number of left turn accidents.

2.2 Augmented Reality HUDs

In recent years, automotive HUDs have been featured as a means of displaying important information to the driver in a manner that does not take their eyes off the road. A commonly cited issue regarding HUDs is the lack of depth to the images being projected or displayed [27]. HUDs traditionally are 2D displays being projected directly onto the windshield or at a fixed distance beyond, resulting in problems with accommodation, vergence, and parallax, where the driver's visual attention is either focused on the HUD or the environment, but not simultaneously. However, advances in display technology and optics have mitigated these potentially dangerous scenarios. Augmented reality (AR) windshields or contact-analog HUDs [20] can display perspective-consistent computer graphic annotations in the driver's field of view that are registered with real elements of road scenery. A

common application scenario is when direct route guidelines from a navigation system are overlaid in the driver's field of view through the windshield [10]. Medenica et al. [16] demonstrated that an AR-based navigation system caused drivers to spend more attention looking ahead on the road than non-AR systems. Frölich et al. [8] varied the realism and viewpoint of route display information and found that for safety-related applications, realistic views provided higher perceived safety than symbolic map views and the more complex the safety recommendation, the more drivers felt realistic visualizations were useful. It is possible that direct annotation of the environment reduces ambiguity of visual instructions involving the driver's environment. A hybrid system that smoothly transitioned between egocentric route information and a 2-D overhead map was simulated in [13] to show the system's potential to reduce navigation errors and reduce driver distraction for elderly drivers. Concepts that may be difficult to perceive, such as speed, can be mapped to alternate spatial representations to better visualize concepts. For example, Tonniss et al. [26] created AR driving aids for longitudinal and lateral driver assistance that visually conforms to the road's surface. AR cues did not interfere with detection of nonhazardous secondary objects and did not impair ability to maintain safe distance behind a lead vehicle. Schall et al. [24] also demonstrated that augmented reality cues helped elderly participants detect hazards without interfering with their primary task of driving. Due to these numerous advantages, AR is a promising modality to visualize spatial relationships between oncoming cars on the road and the driver.

3. LEFT TURN DRIVING AID DEVELOPMENT AND DESIGN

The goal of this work was to develop a system that could help drivers make safer turning decisions and hence, reduce the number of left turn collisions. The system introduced here is an AR-based left-turn driving aid which interprets the velocity of oncoming vehicles as projected paths that indicate where the vehicle will be several seconds in the future. We discuss our methodology and processes that led us to this design in the following section.

3.1 Early Concept: Decision Making

Early concepts focused on drawing a left turn path or guide for the driver only (Figure 1). However, in this concept the arrow puts the driver's focus on the turning path and draws attention away from oncoming vehicles, which are the main source of danger. In addition, displaying a left-turn arrow path relinquishes the left-turn decision-making to the car's control system rather than the driver. From Ragland et al. [21], it is clear that individual drivers have different gap acceptance preferences. Ideally, a computer system would consider the personal preferences of the driver in its decision making. However, we hypothesize that if the computer's suggested left-turn decision does not coincide with the driver's, the driver will fail to trust the capabilities of the system and will eventually ignore it. Building an appropriate level of trust between the driver and the assistive system relies on the driver being aware of the capabilities of the system [15] and communication of the reasoning behind the system's decision making process [18]. The turn arrow concept did not convey the system's strategies to de-



Figure 1: Early left-turn prototype showing turn arrow direction

termine when it was safe to turn left and required the driver to wholly trust the aid, which could result in inappropriate levels of trust.

Another approach to assisting the driver that relies less on the trust between the driver and the system is to improve the driver's model of the situation, rather than tell them what to do. If we examine how a driver makes decisions, a driver's ability to correctly assess the environmental state relies on their situational awareness. *Situation awareness* is used by the driver to understand the current and future state of elements in their operating environment. Endsley [7] looked at situation awareness from an individual's perspective that comprises three levels: I) Perception of elements in the environment, II) Comprehension of their meaning and III) Projection of future system states. The majority of driving aids focus on level I problems such as highlighting pedestrians in a moving vehicle [9], but we believe our visual medium can be used to show future state information to help drivers achieve level II and III awareness. As such, the left turn aid that was developed attempts to project future states of oncoming vehicles to help drivers estimate the correct gap acceptance for turns.

3.2 Proposed Left Turn Aid

Based on the observation that drivers accept smaller gaps for faster oncoming vehicles than slower ones, drivers tend to incorporate relative position and omit speed of vehicles into their decision-making [29]. The proposed left turn aid attempts to make the concept of speed and the future state of the oncoming vehicle easily interpreted as a visual *projected path* (Figure 2). The length of the path is directly computed from the current speed of the oncoming vehicle along with a design variable n indicating the number of seconds we would like to project into the future. In our study, we selected $n = 3$ seconds, based on the second-most frequently selected gap acceptances in observed drivers [21]. The most frequent gap was 2 seconds, but it was also the shortest and least conservative time.

The projected path increases visual saliency of the gap and its length is proportional to a vehicle's speed. The use of red with its negative connotations in traffic contexts provides a visual deterrent for the driver to cross-over the projected

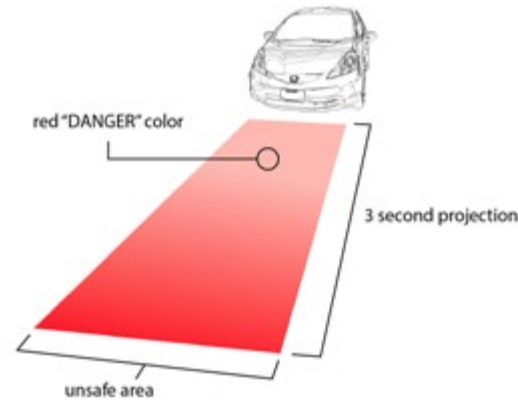


Figure 2: Left-turn projected path driving aid concept

path.

3.3 Qualitative Feedback

With the objective of satisfying user needs and requirements, early involvement of users in the design process can help designers avoid costly formal evaluation and redesign at later stages of development. In this section, we describe our process of selecting and refining the projected path design with early feedback.

Augmented reality applications are designed to enhance the environment to assist the user, but they can also divert the user's visual attention away from other important events occurring in the background, depending on how complex the visual scenery is and how reliable the system is [28, 30]. For the left turn aid, the goal was to design a salient cue that did not occlude the environment heavily so that the driver could still be aware of the other important elements in the scene. Three different visual styles were designed to portray the projected path (Figure 3) under varying levels of occlusion. They included a *solid path*, *chevron path*, and *wireframe path*. The solid path, while easy to detect, covered much of the road, while on the other end of the spectrum the wireframe design was not as visible but did not occlude the scenery heavily. The chevron path represented a compromise, where the chevron was more noticeable than the wireframe, but did not occlude the road to the same extent as the solid path. We did not know whether drivers would respond differently to the various styles or if they preferred one over another. To decide upon a projected path visual style, we completed a qualitative study to judge user reactions and preferences.

Each participant was asked to view 3 videos of the same left turn scenario but with different projected path visual styles overlaid. The participants were free to replay these videos as many times as they wanted and were then asked to fill out a questionnaire. The survey questions were aimed at capturing user opinion on aspects important for the left turn aid including:

- Whether or not the projected path was difficult to see



Figure 3: Different projected path visual styles for early user feedback. The chevron is green to increase visual saliency. A red chevron was difficult to see.

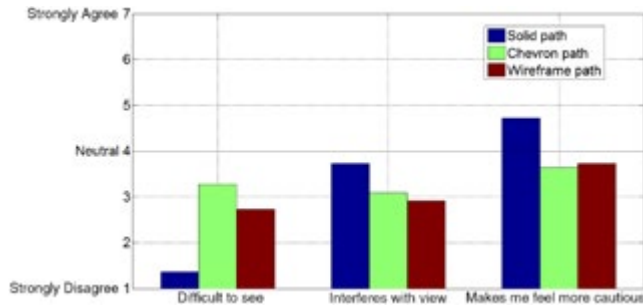


Figure 4: Early user feedback on 3 selected projected path visual styles

(visibility)

- Whether or not the projected path interfered with the driver's ability to assess the environment
- Whether or not the projected path made the driver feel more cautious when making a left turn

Participants responded to each question using a 7 point Likert scale where 1 = strongly disagree, and 7 = strongly agree. In addition, participants' understanding of the overlaid visuals, their subjective preference and suggestions for possible improvements were recorded.

11 participants (8 males and 3 females) with the ages ranging from 26 to 40 completed the evaluation. Figure 4 shows outputs of the scored survey questions. Compared to the chevron and wireframe path, the solid path appeared to have more distinguished characteristics. It was very easy to see and appeared to make the driver slightly more cautious. Though the solid path had higher interference ratings compared to the other visual styles, participants did not think it interfered with the environment overall. In the general comments, we found that participants roughly understood the meaning of all 3 projected path styles. More subjects liked the appearance of the chevron path the most because it obstructed their view of the road less and one participant noted that it gave a sensation of motion. However, it was also noted that the chevron path seemed to cause more eye movements (moving back and forth between the chevron

path and the actual vehicle) and could be a bit distracting, making the participant feel dizzy. Based on both the output of scored questions and open-ended comments, we decide that the solid projected path was best suited for the left turn aid since it is easier to see and it makes the driver a bit more cautious, which are both important goals of the left turn aid. Taking into account the users comments, we reduced the transparency of the solid path to make it less obstructive to the driver's view.

4. PROTOTYPE SYSTEM OVERVIEW

Our prototype system (Figure 5) consists of a curved 120 degree projection screen, the National Advanced Driving Simulator (NADS) MiniSim software for driving simulation, and a 3D AR-HUD. Due to the large projection area, two projectors are needed to cover the entire screen, with blending software employed two seamlessly blend the two projection areas into a single large display area. Since the left-turn aid is employed at controlled intersections, the large field of view of the projector was necessary to capture a sufficiently immersive driving experience while turning and stopping at the road intersection.

In front of the projection screen is a 3D combiner-HUD prototype that is capable of generating images at different focal depths, reducing the safety issues related to focal distance. The head position of the driver is located 5m in front of the projection screen. Consequently, due to the fixed distance between the HUD and screen, we set the focal depth of augmented graphics in our HUD to the same depth of 5m for simulation. In a real environment, the focal depth would vary along the projected path. The head of the driver must reside within a finite volume region of about 20 cm x 10 cm x 10 cm in front of the windshield glass called the *eye-box*. The center of the eye box is located approximately 70 cm from the HUD. If the head moves outside of this eye-box, the image in the windshield disappears from view. This restriction is not unreasonable as the drivers tend to keep their head position steady while driving.

The HUD computer generates the augmented graphics to the HUD display. The driver sits in a chair that can be manually adjusted to position the driver's head in the eye-box. The HUD computer hosts the left-turn driving aid which determines how to generate the projected path based on oncoming car positions and the presence of road inter-

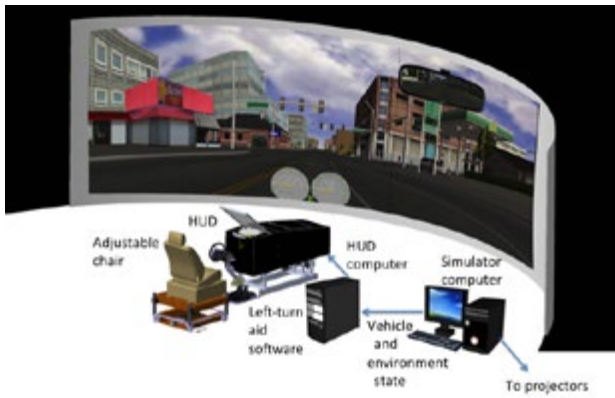


Figure 5: Diagram showing prototype setup. Participants sit in front of a 3D AR-HUD which is placed in front of a curved, 120 degree projection screen.

sections. This environmental information is provided by the driving simulator computer over a networked socket connection during the simulation session. A video camera records the entire session for later analysis.

5. PILOT STUDY

A pilot study was developed to study the effects of the projected path and to help determine the best evaluation methods for a more formal usability study. The pilot described here was designed to test whether or not the presence of the projected path influenced when the driver decided to make a left turn, and what (if any) effects it had on driver behavior. Specifically, we were interested in observing the gap acceptance decision, which is the time gap between oncoming vehicles when the driver decides to complete the left turn. Because the length of the projected path is dependent on the oncoming vehicle's speed, we were also interested in observing whether the projected path would have different effects at high and low oncoming vehicle speeds. The proposed study tests the driver's left turn behavior at urban intersections with vehicles approaching at both 35 and 55 mph. Performing the pilot study also provided us with some insights regarding the driving aid and these observations are outlined in the discussion.

5.1 Methods

To test the effects of the projected path, drivers were instructed to complete a driving course under two conditions: with the driving aid and without. The details of the course and procedures are provided below.

5.1.1 Traffic Scenario

Two driving test courses were designed to guide participants through a series of left turns at signaled intersections. Each intersection consisted of a 4 or 2 lane divided road (see Figure 6 for the 4 lane example). As the participant approached the intersection, a series of oncoming vehicles would begin to enter the intersection in the near lane at increasing time gap intervals. In order to measure the driver's minimum gap acceptance, the time gaps increased linearly from 1 to 8 seconds, in 1 second increments. A maximum time gap of 8 seconds was chosen based on previous gap acceptance

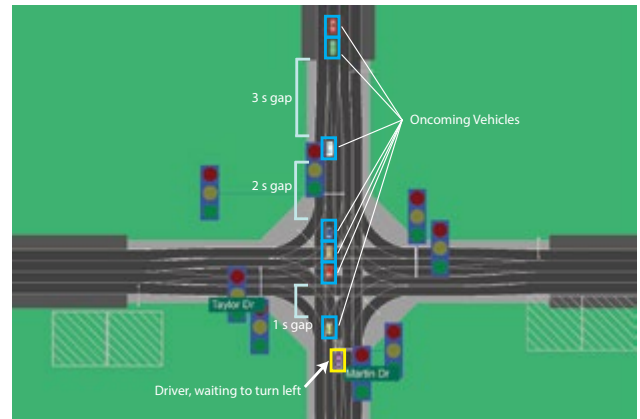


Figure 6: Diagram of oncoming traffic scenario at each intersection. As the driver approaches the intersection, vehicles approach with linearly increasing time gaps. Random vehicles with short time gaps are interspersed between to randomize the traffic flow.

findings [29]. To help the traffic appear random and prevent participants from learning the time gap patterns, in between vehicles with the specified time gaps, a separate sequence of 1 to 3 vehicles with very small gaps (1-2 seconds) was inserted. The number of vehicles inserted in-between was randomized at each intersection to help make the traffic semi-irregular, and to prevent participants from counting the number of vehicles passing by to decide when to turn left. The design of this traffic sequence was inspired by previous work by [29]. The traffic was timed to begin approaching the intersection at the same time as the driver.

Each test course simulated an urban environment that consisted of 10 intersections during which the participant was instructed to make a left turn. Two different speeds of the oncoming vehicles were tested. At half of the intersections, the speed of the vehicle was 35 mph, and for the remaining intersections, the speed was 55 mph. The order of the speeds was randomized throughout the course so that the driver could not anticipate the speeds of the vehicles. The two speeds were chosen to observe driving behavior at a low and relatively high speed. In urban settings, the oncoming speed is likely to be within these limits. The speed of the oncoming vehicles was also kept constant as they passed through the intersection.

5.1.2 Procedure

Participants initially completed a demographic questionnaire that also recorded driving experience, recent accidents, whether or not they had a history of motion sickness, or experience with augmented reality applications. Then, the participants were introduced to the driving simulator and HUD setup, where the experimenter explained the details of how to operate the driving simulator and familiarized them with the digital instrument panel display and HUD. An eye gaze tracker was also placed on the subject and calibrated to record their gaze patterns during the experiment. A training course with 4 different left turn intersections was then completed to allow



Figure 7: Screenshots of two driving conditions. (a) shows the condition without the driving aid (no projected path) and (b) displays the condition with the driving aid (with projected path).

participants to become accustomed to driving in the simulator environment (using the accelerator and brake pedals, steering wheel) and to practice turning left and understand the main task. Once the training session was complete, subjects proceeded to the experiment phase.

In the main experiment, subjects were asked to complete the two driving courses, once with the projected path and once without the projected path (see Figure 7). To reduce presentation order bias, the order of the two conditions (without driving aid, with driving aid) was randomized between subjects. The driving course associated with each condition was also randomized between subjects to further reduce potential biases. Subjects were instructed to stop at any point during the experiment if any motion sickness became apparent. They were allowed to take a break in between driving courses if desired. Each driving course took approximately 10 minutes to complete. The experimenter also recorded video of the experiment for post-study analysis.

After finishing both driving courses, the participant was asked to complete a qualitative survey that was designed to capture their opinions on the driving aid and offer any comments or suggestions regarding the experiment.

5.1.3 Analysis

Although use of the driving simulator allows a wide range of variables to be measured, in this pilot study we were mainly interested in the gap acceptance. The gap acceptance is traditionally defined as the time between the rear bumper and front bumper of two oncoming vehicles when the driver decides to make a turn. Errors in automated simulator mea-

surements led us to rely on the video recordings of the experiment to estimate the gap acceptance in this pilot study. Because in the video recordings, the time gap between vehicles when the driver completes the left turn is not possible (the second vehicle is out of the field of view as the driver turns the corner), in the following analysis, the gap acceptance is based on the measured time gap between vehicles prior to the driver initiating the left turn. Because the time gap increases linearly in 1 second increments, we estimate the gap acceptance to be the time gap prior to turning + 1 second.

5.2 Results

Four pilot subjects completed the study successfully. Their ages ranged from 26 to 32 (mean = 29.75), with 3 males and 1 female, and an average of 12.25 years of driving experience. Three other participants withdraw due to motion sickness. Due to the small sample size of the pilot study, only trends and general observations are discussed.

5.2.1 Gap Acceptance

Initial data indicate that changes in driver gap acceptance between the two main conditions vary from subject to subject. Although trends seem to indicate that drivers appeared to be slightly more conservative with their gap acceptance with the projected path, individual subjects performed differently with the aid and without the driving aid. Two subjects were more conservative with their gap acceptances, while one subject was more aggressive in the presence of the driving aid. The fourth participant showed no difference in gap acceptance with the aid.

Similar trends were found when breaking the data down according to the varying speeds of the oncoming vehicles. Overall, participants tended to choose larger acceptance gaps when the oncoming vehicle speed was lower (35 mph). This is consistent with other left-turn gap acceptance studies [29].

6. DISCUSSION

Although the sample size from this pilot study is too small to draw any definitive conclusions, we gained several insights that have implications on the design of the driving aid and how to test its effectiveness.

6.1 Driving Attitudes

We observed that different participants approached the driving course with varying behaviors, where some were more aggressive than others. In the case of the more aggressive driver, the presence of the projected path appeared to increase their gap acceptance, whereas for the more conservative driver, the gap acceptance decreased with the projected path. It is possible that driver attitudes will influence the way in which the driving aid is interpreted. For the more conservative driver who overcompensates for gap acceptance, the presence of the aid may increase their confidence in knowing when it is safe to turn, whereas for the more aggressive driver, the aid may be more helpful in illuminating a potentially dangerous scenario. In future studies, driver behavior and attitudes will also be recorded to partition subjects into different categories based on driving style. Questionnaires based on testing Type A and Type B per-

sonality tendencies [12] can help detect a driver's inclination to be aggressive.

6.2 Projected Path User Feedback

Although the projected path was clearly visible, the user comments indicate that improvements to the design of the projected path driving aid should be considered. Some participants indicated that the meaning of the projected path was not clear and found it distracting. The projected path also did not help them feel more confident that the oncoming vehicle would not crash with them, and the path required more cognitive effort to process. Though users may prefer a more subtle cue, the presence of the information may still lead to safer driving habits. A similar case is the implementation of roundabouts. In surveys, a large percentage of drivers (55%) did not favor the inclusion of roundabouts at intersections [23]; however, studies have shown that roundabouts decrease traffic accidents [19]. One theory is that because the roundabout requires more cognitive effort to interpret, driver's attention is heightened. This may be the case with the projected path but we must find a balance between subtlety and effectiveness. Modifications to the projected path visual attributes, such as color, shape, transparency, should be further iterated upon to design a more user acceptable driving aid with a good visceral response to using it.

The current length of the projected path indicates the space the oncoming vehicle will occupy in the next 3 seconds. However, the average gap acceptance for the different participants varied from 4 to 6 seconds. One reason why the difference in gap acceptance for the two conditions was small may have been because the average gap acceptance already exceeded the 3 second path length. The length of the projected path has not been optimized and it is possible that extending the path to cover 4 or 5 seconds into the future would impact the driver's gap acceptance even more. In future studies, we plan to explore the path length as an additional factor in our experiment to determine whether or not behavior changes

6.3 Simulator Motion Sickness

Simulator motion sickness is a known problem for driving simulators, with several studies documenting the phenomena. A high percentage of our pilot subjects withdrew from the study due to feeling nauseated or dizzy from the driving scenario. The study required participants to brake and turn numerous times over the course of 20 minutes, and in [29], turning and braking maneuvers were observed to cause motion sickness. In hindsight, testing the aid through full simulation was not an ideal method and alternative methods to evaluate the aid should be explored. For example, evaluation of the gap acceptance behavior may be obtained by asking participants to indicate when they would turn, but not complete the turn. The driving scenario may also be modified to include fewer buildings and objects, which move across the screen during the turn and may contribute to the participant's dizziness.

7. CONCLUSIONS

In this paper, we present a left turn driving aid for use with a volumetric 3D Head-Up-Display with preliminary observations from an early stage pilot test. Utilizing the advantages

of a 3D AR-HUD, the driving aid interprets the velocity of an oncoming vehicle as a 3 second projected path that is overlaid on the road in front of the vehicle. The goal of the projected path is to help drivers determine if it is safe to execute a left turn by indicating where the oncoming vehicle is expected to be in the future. Although the small sample size of the pilot study does not allow for definitive findings, valuable insights on how to modify the aid and future evaluations were gained.

When presenting users with different visual style options for the projected path, we found that a solid red bar provided the most salient cue and made users feel more cautious. In pilot tests, we found that driver attitudes were an additional factor to consider and that different drivers may prefer different lengths of the projected path. Future work will incorporate testing these additional factors and conditions to optimize the design of the turning aid and its effectiveness. Simulator discomfort was heightened in our pilot study due to the numerous turns and braking maneuvers required. Modifications to the testing course are planned to minimize user discomfort. We may also employ simpler testing techniques that allow users to indicate when they would make the turn without having to complete the maneuver. Although the projected path appears to increase the cognitive effort of the drivers, whether this has a negative impact on driving behavior is unknown. Additional formal evaluation with a larger, more diverse subject pool must be completed to understand the potential of this application.

8. ACKNOWLEDGMENTS

The authors wish to thank Tom Zamojdo and Chris Grabowski for their stimulating conversations, sage advice, and assistance in building our HUD prototypes, and Lee Beckwith for his contributions to the design of the driving aid.

9. REFERENCES

- [1] M. Ablabmeier, T. Poitschke, F. Wallhoff, K. Bengler, and G. Rigoll. Eye Gaze Studies Comparing Head-up and Head-down Displays in Vehicles. *IEEE International Conference on Multimedia and Expo*, pages 2250–2252, 2007.
- [2] R.T. Azuma. A Survey of Augmented Reality. *Presence*, 6(4):355–385, 1997.
- [3] J. Caird and P. Hancock. The perception of arrival time for different oncoming vehicles at an intersection. *Ecological Psychology*, 6(2):83–109, 1994.
- [4] J.K. Caird and P.A. Hancock. Left-turn and Gap Acceptance Crashes. *Human Factors in Traffic Safety*, pages 613–652, 2002.
- [5] E. Choi. Crash factors in intersection-related crashes: An on-scene perspective. Technical Report DOT HS 811 366, Mathematical Analysis Division, National Center for Statistics and Analysis, National Highway Traffic Safety Administration, 2010.
- [6] J. Chovan, L. Tijerina, J. Everson, J. Pierowicz, and D. Hendricks. Examination of intersection, left turn across path crashes and potential IVHS countermeasures. Technical Report DOT-VNTSC-NHTSA-94-4, U.S. Department of Transportation, September 1994.
- [7] M. Endsly. Towards a theory of situation awareness in

- dynamic systems. *Human Factors*, 37(1):32–64, 1995.
- [8] P. Fröhlich, R. Schatz, P. Leitner, M. Baldauf, and S. Mantler. Augmenting the driver's view with realtime safety-related information. In *Proceedings of the 1st Augmented Human International Conference*, pages 1–11, 2010.
- [9] D. Gavrilu and S. Munder. Multi-cue pedestrian detection and tracking from a moving vehicle. *International Journal of Computer Vision*, 73(1):41–59, 2007.
- [10] D. Harkin, W. Cartwright, and M. Black. Decomposing the map: using head-up display for vehicle navigation. In *Proceedings of the 21st International Cartographic Conference*, 2005.
- [11] IBEO. Ibeo Automotive Systems. <http://ibeo-as.com/index.php/en/iobject-tracking>, 2012.
- [12] C. Jenkins, S. Zyzanski, and R. Rosenman. Progress toward validation of computer-scored test for the type a coronary-prone behavior pattern. *Psychosomatic Medicine*, 33:193–202, 1971.
- [13] S. Kim and A. Dey. Simulated augmented reality windshield display as a cognitive mapping aid for elder driver navigation. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, pages 133–142, 2009.
- [14] T. Lajunen, A. Corry, H. Summala, and L. Hartley. Cross-cultural differences in drivers' self-assessments of their perceptual-motor and safety skills: Australians and finns. *Personal. Individual Differ.*, 24:539–550, 1998.
- [15] John D. Lee, Katrina A. See, and Iowa City. Trust in automation: Designing for appropriate reliance. *Human Factors*, 46:50–80, 2004.
- [16] Zeljko Medenica, Andrew L. Kun, Tim Paek, and Oskar Palinko. Augmented reality vs. street views: a driving simulator study comparing two emerging navigation aids. In *Proceedings of the 13th International Conference on Human Computer Interaction with Mobile Devices and Services*, pages 265–274, 2011.
- [17] J. Misener and et al. Cooperative Intersection Collision Avoidance System (CICAS): Signalized Left Turn Assist and Traffic Signal Adaptation. *California PATH Research Report*, 2010.
- [18] D. Norman. *The Design of Future Things*. Basic Books, 2007.
- [19] Bhagwant Naraine Persaud, Richard A Retting, Per E Garder, and Dominique Lord. *Crash reductions following installation of roundabouts in the United States*. Insurance Institute for Highway Safety, 2000.
- [20] T. Poitschke, M. Ablassmeier, G. Rigoll, S. Bardins, S. Kohlbecher, and E. Schneider. Contact-analog information representation in an automotive head-up display. In *Proceedings of the 2008 symposium on Eye tracking research & applications*, pages 119–122, 2008.
- [21] D. Ragland, S. Arroyo, S. Shladover, J. Misener, and C. Chan. Gap acceptance for vehicles turning left across on-coming traffic: Implications for intersection decision support design. Technical Report UCB-ITS-TSC-2006-2, UC Berkeley, April 2006.
- [22] T.A. Ranney, W.R. Garrott, and M.J. Goodman. NHTSA Driver Distraction Research: Past, Present, and Future. *NHTSA Research and Development, Washington, DC*, 2000.
- [23] R.A. Retting, G. Luttrell, E.R. Russell, and Insurance Institute for Highway Safety. *Public Opinion and Traffic Flow Impacts of Newly Installed Modern Roundabouts in the United States*. Insurance Institute for Highway Safety, 2001.
- [24] Mark C. Schall, Michelle L. Rusch, John D. Lee, Jeffrey D. Dawson, Geb Thomas, Nazan Aksan, and Matthew Rizzo. Augmented reality cues and elderly driver hazard perception. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 55(3):643–658, 2013.
- [25] Z. Sun, G. Bebis, and R. Miller. On-Road Vehicle Detection: A Review. *IEEE Trans. on Pattern Analysis and Machine Intelligence*, 28(5):694–711, 2006.
- [26] M. Tönnis, C. Lange, and G. Klinker. Visual longitudinal and lateral driving assistance in the head-up display of cars. In *In Proceedings of the 6th International Symposium on Mixed and Augmented Reality (ISMAR)*, 2007.
- [27] D. R. Tufano. Automotive huds: the overlooked safety issues. *Human factors*, 39(2):303–311, 1997.
- [28] N. J. Ward, A. Parkes, and P. R. Crone. Effect of background scene complexity and field dependence on the legibility of head-up displays for automotive applications. *Human factors*, 37(4):735–745, 1995.
- [29] X. Yan, E. Radwan, and D. Guo. Effects of major-road vehicle speed and driver age and gender on left-turn gap acceptance. *Accident Analysis and Prevention*, 39:843–852, 2007.
- [30] Michelle Yeh and Christopher D. Wickens. Display signaling in augmented reality: Effects of cue reliability and image realism on attention allocation and trust calibration. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, 43(3):355–365, 2001.